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Computing Project

PIG: GPU Redesign of a Plenoptic Imaging Pipeline and Extension to Multi-View Reconstruction

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Abstract

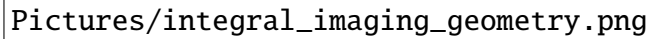
This report presents the redesign of an existing CPU plenoptic image generation (PIG) pipeline toward a GPU implementation using CUDA. This work focuses on the acceleration about the most computationally expensive stages of the existing pipeline, and in the same time aims to preserve the rendering correctness. In addition, the new GPU pipeline is extended to support multi-view datasets (RGB + depth) by means GPU-based point cloud registration and merging. Experimental results demonstrate significant runtime improvements and validate the geometric consistency and efficiency of the proposed multi-view pipeline regarding the single-view disocclusion problems.

Background Context and Motivation

1.1 Integral Imaging Fundamentals

Integral Imaging (InI) is a glasses-free three-dimensional visualization technique based on light-field reconstruction through a Microlens Array (MLA). A plenoptic image, composed of a grid of microimages, is displayed behind the MLA, allowing each microlens to project a slightly different view of the scene toward the observer. This enables depth perception and motion parallax without requiring wearable devices. [Fer+26]

The display geometry is mainly defined by the MLA pitch D , focal length f , and the gap B between the MLA and the display panel. These parameters determine the reconstruction properties of the system. The following figure (adapted from the figure 2 in [Fer+26]) explains better the setup of the Integral Imaging.



Pictures/integral_imaging_geometry.png

Figure 1.1: Geometric model of an Integral Imaging display system. Adapted from [Fer+26].

The reconstructed image is focused around the Center Depth Plane (CDP), whose position is derived from the thin lens equation:

$$l = \frac{Bf}{B - f} \quad (1.1)$$

The existing software adopted uses a square MLA configuration with a pitch of 1.6056 mm, focal length of 8.028 mm, and a display–MLA gap configured for real-mode operation.

1.2 Existing Software

The Plenoptic Image Generator (PIG) software converts RGB+D data into a plenoptic image suitable for Integral Imaging displays. Basically a plenoptic image is 2D image composed by many microimages, each of these associated to a microlens. The magic happens when the plenoptic image is shown behind the MLA: each microlens projects their own microimage observed from a viewpoint slightly different onto the space, the light field is reconstructed and the observer perceives 3D effects.

Therefore at the end of the day PIG is able to convert data RGB+D from the initial scene, into microimages that are supposed to be seen through each microlens. Technically speaking PIG is entirely CPU-based and organized into 4 stages: Pre Processing, Point Cloud Reconstruction, Plenoptic Rendering and Post Processing.

The Pre-Processing stage applies super-resolution upscaling to RGB and depth images in order to densify the reconstructed point cloud. In fact a point cloud generated by only one RGB+D is sparse so the paper contributors have shown “crack artifacts” (holes) occur.

The Point Cloud Reconstruction stage converts RGD+D data into a cloud of 3D points by taking into account the intrinsic camera parameters. Assuming a pinhole camera model, each pixel (x_i, y_i) associated with depth Z_i is backprojected into 3D space according to:

$$(X_i, Y_i, Z_i)^T = \left(\frac{(x_i - p_x)Z_i}{f_c}, \frac{(y_i - p_y)Z_i}{f_c}, Z_i \right)^T \quad (1.2)$$

where f_c denotes the camera focal length and (p_x, p_y) the principal point coordinates.

Now with the Plenoptic Rendering stage we need to figure out for each 3D point from the previous stage which microlens are able to see it and where such point appears inside each microimage.

They used the center coordinates of the center of each microlens together with its angular Field-of-View (see equation 8 in [Fer+26]). Once the microlens has been found the point is projected into the relative microimage accordingly to

$$\begin{aligned} x'_{ij} &= (X_i - m_{jx}) \frac{B}{Z_i} + m_{jx} \\ y'_{ij} &= (Y_i - m_{jy}) \frac{B}{Z_i} + m_{jy} \end{aligned} \quad (1.3)$$

where m_{jx} and m_{jy} are the center coordinates of the microlens. However here they observed as well that when $Z_i \rightarrow 0$ the projection

$$\frac{B}{Z_i}$$

diverges. In other words the points near to the MLA explode out of the microimage causing artifacts. Contributors resolved this problem by filtering in the point cloud only the points above a certain depth threshold.

The last stage is about filling the remaining “crack artifacts”. Even if they applied an up-scaling via super-resolution some pixels remain empty due to the discrete nature of the point cloud. In order to address such inconvenience, once identified invalid pixel by using $\alpha=0$, they applied local median filter and copied only filtered values into holes.

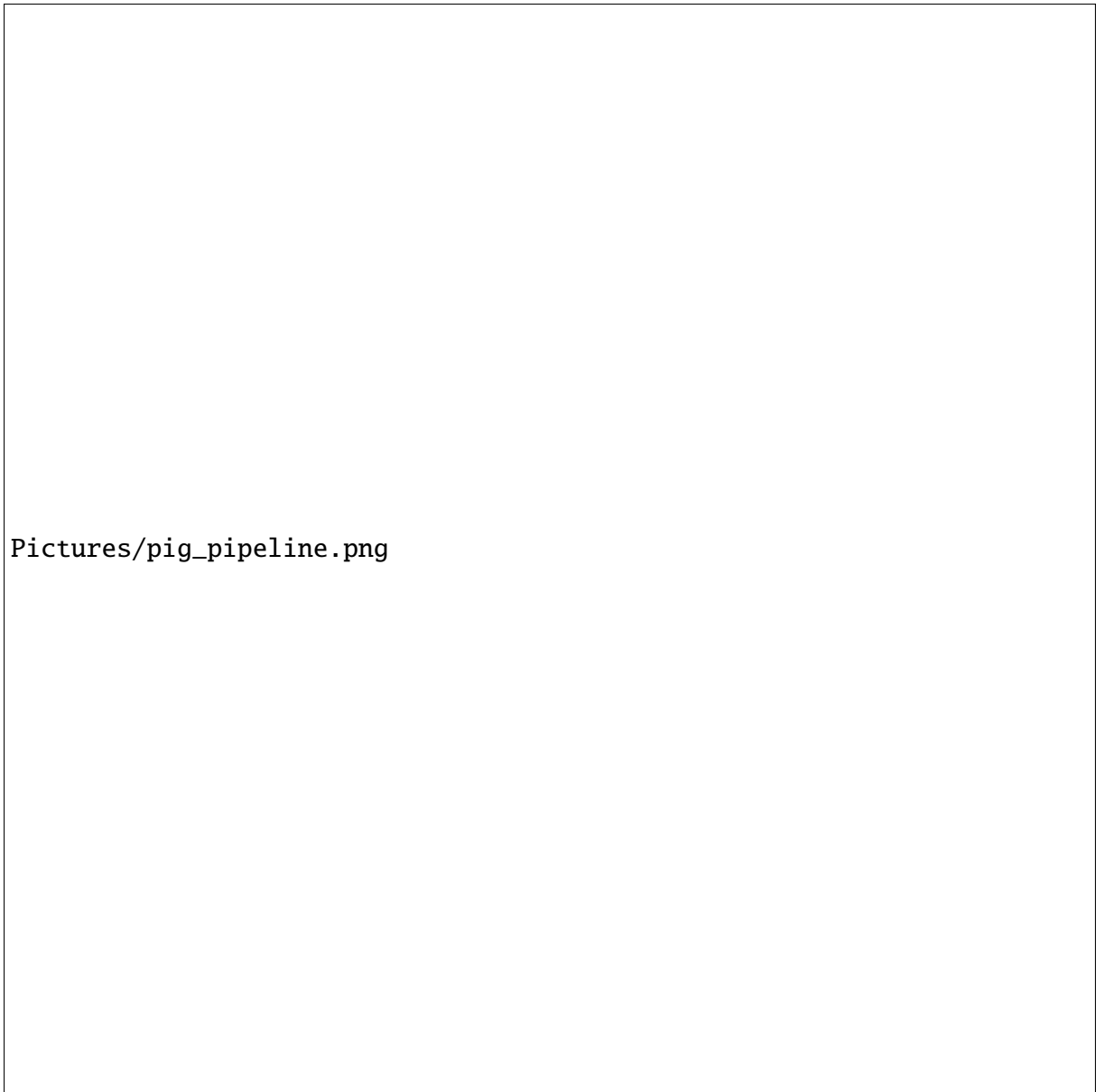


Figure 1.2: Overview of the original PIG rendering pipeline. Adapted from [Fer+26].

1.3 Project Objectives

The existing CPU-based pipeline works but suffers from runtime performances, due to its sequential nature. Stages like Point Cloud Reconstruction and Post Processing involve large amount of pixel-wise and point-wise operations making them computationally expensive. Moreover contributors observed disocclusion artifacts in the plenoptic image generated. This because of the originate scene comes from a single RGB+D viewpoint and in such scenario regions occluded cannot be recovered during the rendering.

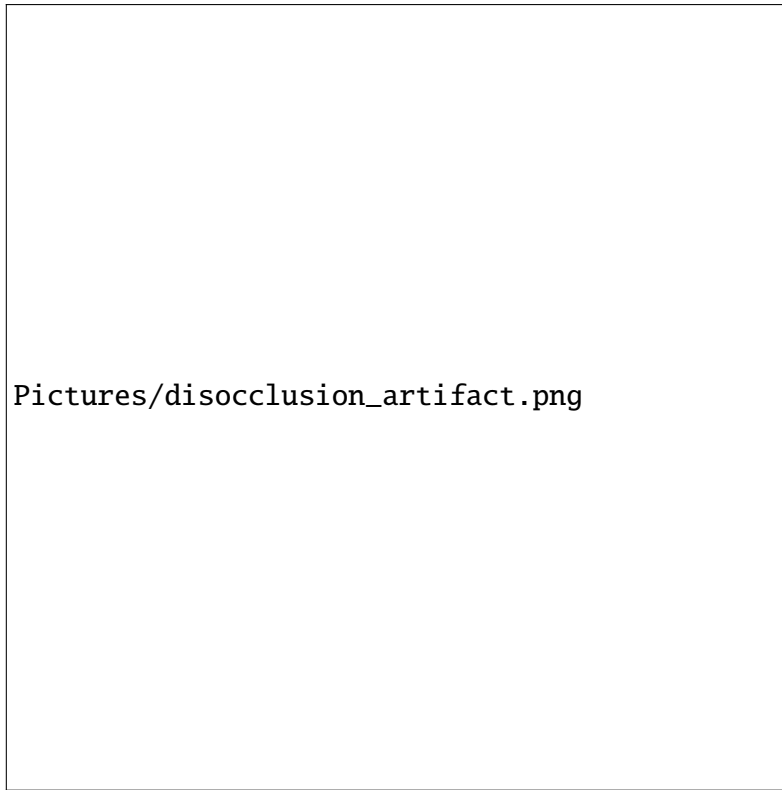


Figure 1.3: Example of crack and disocclusion artifacts generated by single-view point-cloud rendering. Adapted from [Fer+26].

These limitations motivate naturally the project objectives of this work. The first one is to redesign the most time computationally expensive stages of the original pipeline into a GPU execution while preserving rendering correctness and visual consistency. The second objective is to extend the new pipeline by implementing a new multi-view reconstruction pipeline by taking into account datasets with multiple RGB+D data.

Bibliography

- [Fer+26] Ferreira, Brenno et al. (2026). “Large-Size Integral Imaging Display with Depth Image-Based Plenoptic Rendering”. In: *SPIE*.